

MODULE-4

Programming tools

Evolution of embedded programming tools,
Modelling programs, Code optimization, Logic
analyzers, Programming environment

UNIFIED Modelling LANGUAGE (UML)

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❑ What is Modelling Language?

A Modelling language is a graphical/textual computer language explains design and construction of any model or structure following a set of guidelines.

❑ Importance of Modelling : Visualization, reduced complexity, documentation

❑ To effectively model a system, you need a language with which the model can be described and here's where UML comes in.

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INTRODUCTION

- ❑ UML - general purpose visual Modelling language
- ❑ It is used to visualize, specify, construct, and document
- ❑ Using tools codes can be generated with UML diagrams in various languages
- ❑ It is a pictorial language
- ❑ It can be used for both Modelling the software system and non-software systems
- ❑ For example, modelling a process flow in automated systems, etc.

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INTRODUCTION

- ❑ UML is used in many applications by many people like business users, common people to make the system simple, clear and understandable
- ❑ It is a simple modelling mechanism to model most of the practical systems
- ❑ It is a object-oriented analysis based method

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INTRODUCTION

UML Uses:

- ☐ Forecast systems
- ☐ To estimate the reusability.
- ☐ Lower costs
- ☐ Plan and analyze system behavior
- ☐ Easier maintenance/modification

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INTRODUCTION

There are two types of diagram in UML

1. **Structure diagram:** explains static structure of a system

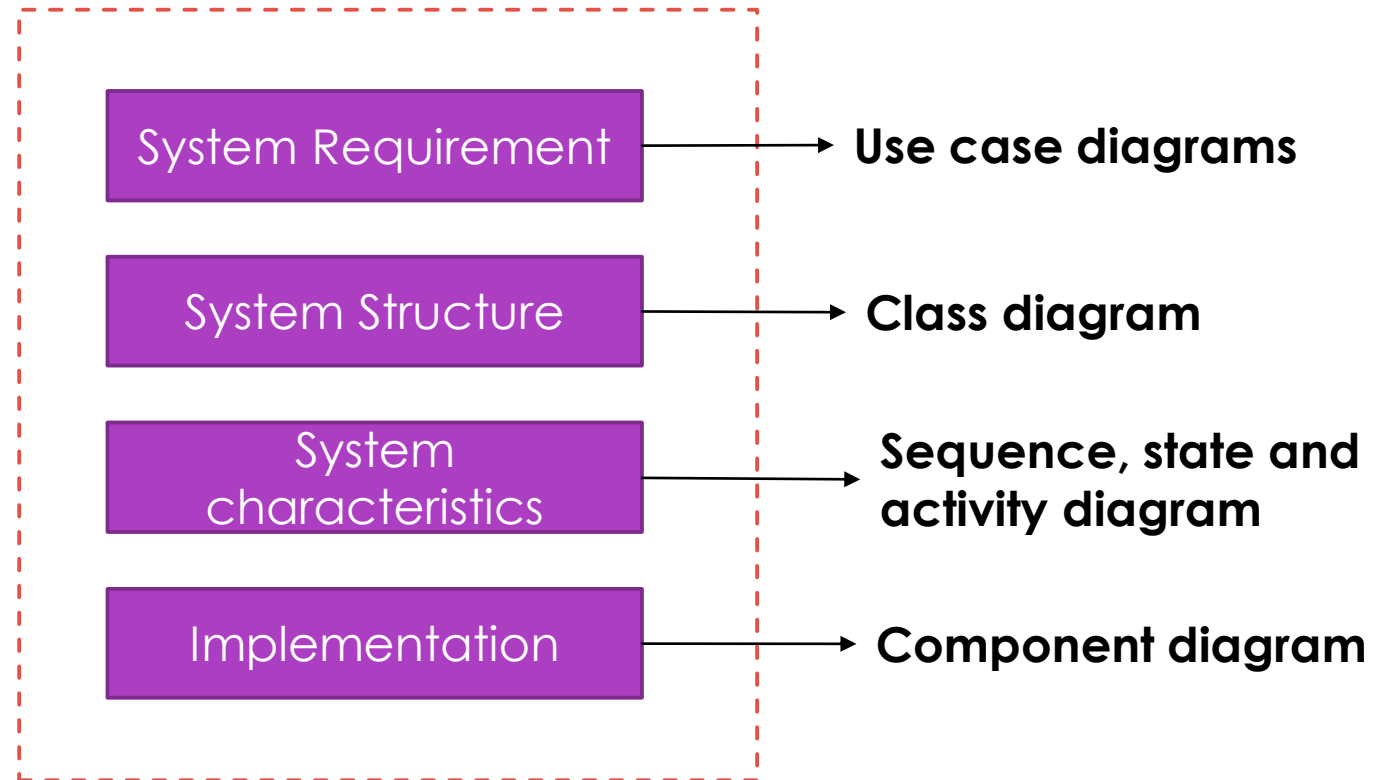
- Class diagram
- Object diagram
- Deployment diagram
- Package diagram

2. **Behavior diagram :** used to model dynamic changes

- Use case diagram
- Interaction diagram
- Activity diagram
- State diagram

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❑ Principles of UML Modelling



USE CASE DIAGRAM

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USE CASE DIAGRAM

- ❑ The use case diagram shows different ways a user can interact with the system
- ❑ It can be used in scenarios like when system interact with people or any other external systems
- ❑ It contains “use-cases” and “actors”
- ❑ Use case diagram depicts how the actor interacts with use cases
- ❑ It describes the relationship between functionalities and their controllers

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USE CASE DIAGRAM

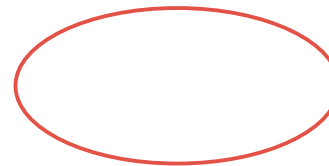
System



Actors



Use Cases



Relationships



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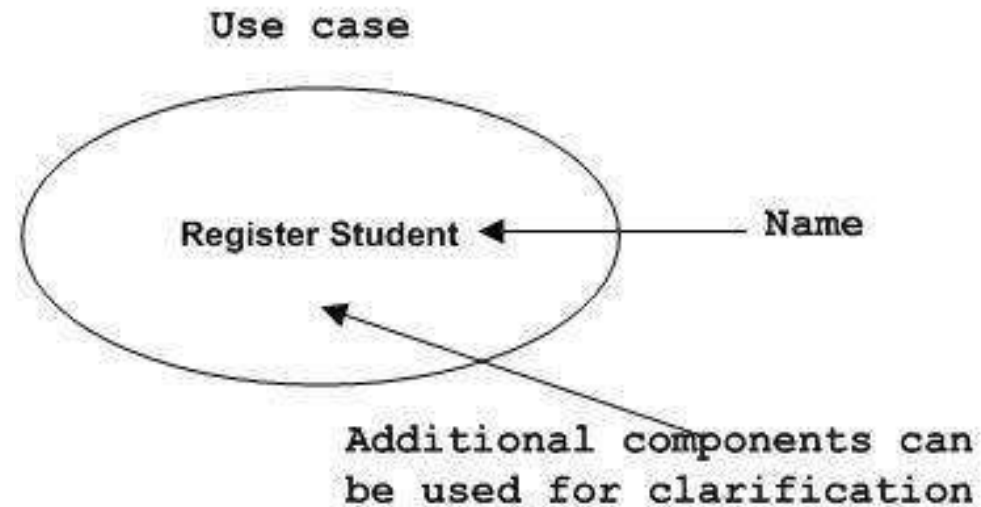
UML NOTATIONS

- ❑ Notations are very important in any modelling language
- ❑ Efficient use of notations play a major role in making any model meaningful
- ❑ Use case diagram: Notations of things and relationships
- ❑ Extensibility makes UML more powerful and flexible.

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USE CASE NOTATIONS

- ❑ Use case - high level functionalities of a system.
- ❑ Use case is represented as an eclipse



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USE CASE NOTATIONS - ACTOR

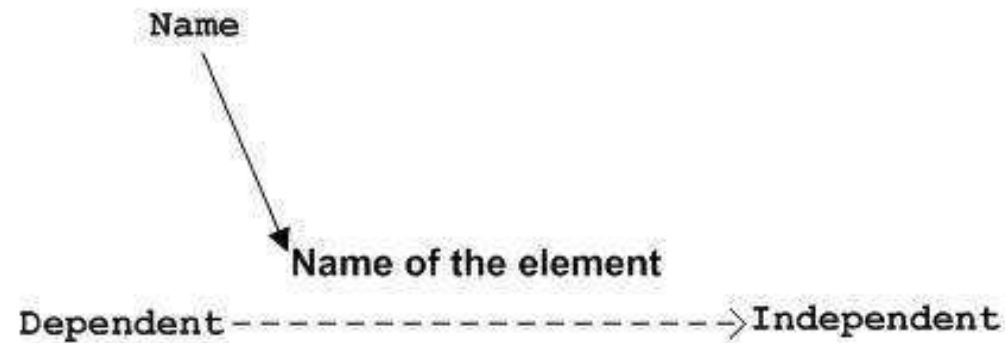
- ❑ An actor - the internal or external entities.



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USE CASE NOTATIONS - DEPENDENCY

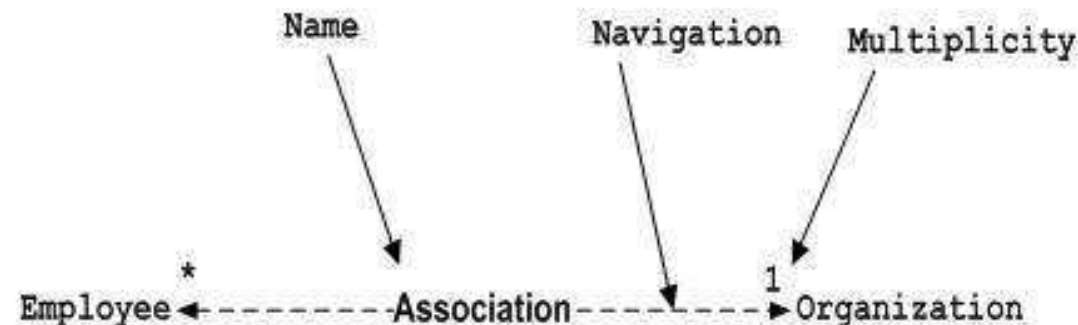
- ❑ Dependency - relationship between two elements of a system
- ❑ Dependency - dotted arrow
- ❑ Arrow head represents - independent element
- ❑ Other end - dependent element



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USE CASE NOTATIONS - ASSOCIATION

- ❑ Association - relationship between two elements
- ❑ Association describes how the elements are associated.
- ❑ Association is represented by a dotted line with (without) arrows on both sides
- ❑ The multiplicity (1, *, etc.) to show how many objects are associated



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USE CASE NOTATIONS - GENERALIZATION

- ❑ Generalization - parent-child relationship
- ❑ Generalization - inheritance relationship of the object-oriented concept
- ❑ Generalization - arrow with a hollow arrow head

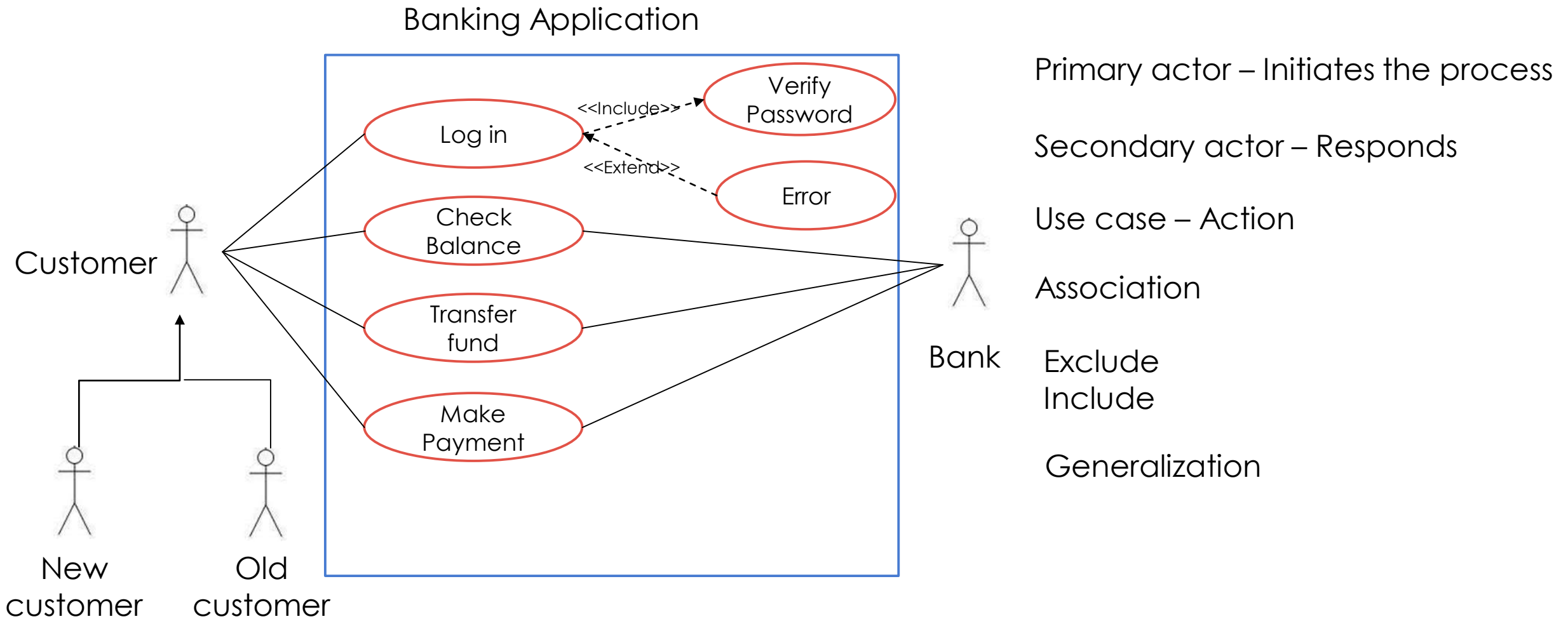


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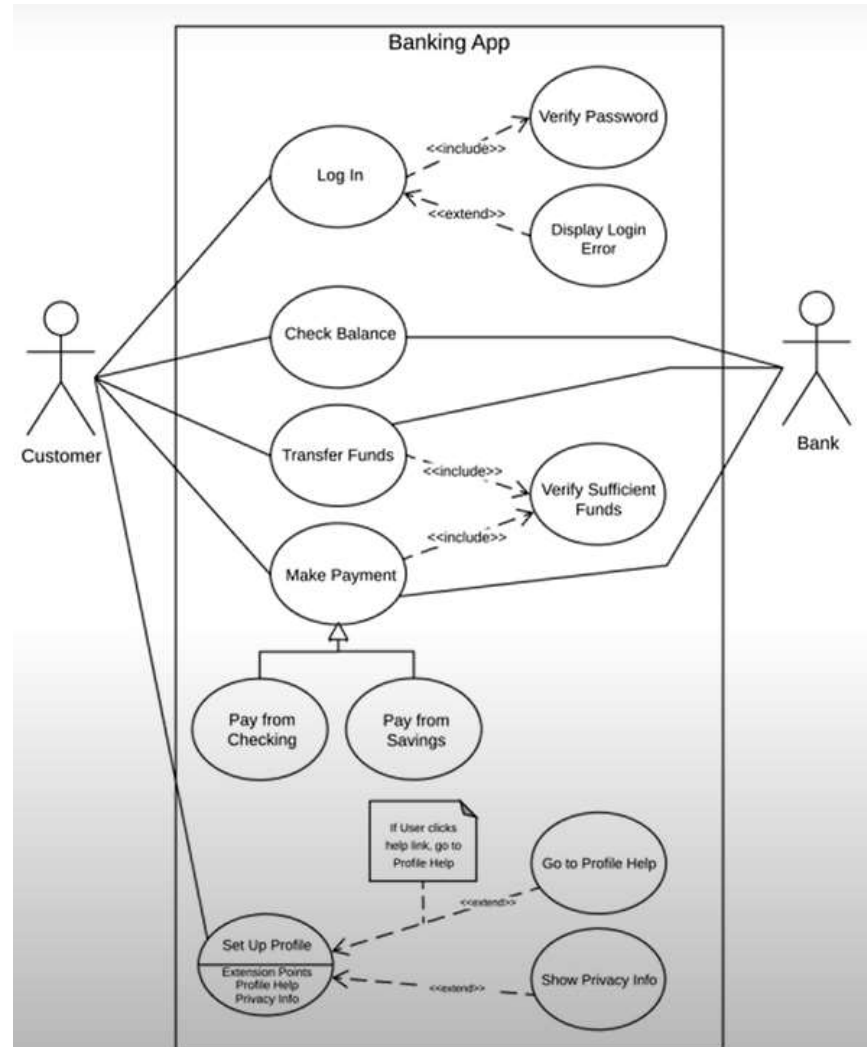
USE CASE NOTATIONS - <<include>> and <<extend>>

- ❑ 'Extend' and 'Include' – between use cases
- ❑ Include: invocation of one use case by the other
- ❑ Extend: extending use case will work exactly like the base use case

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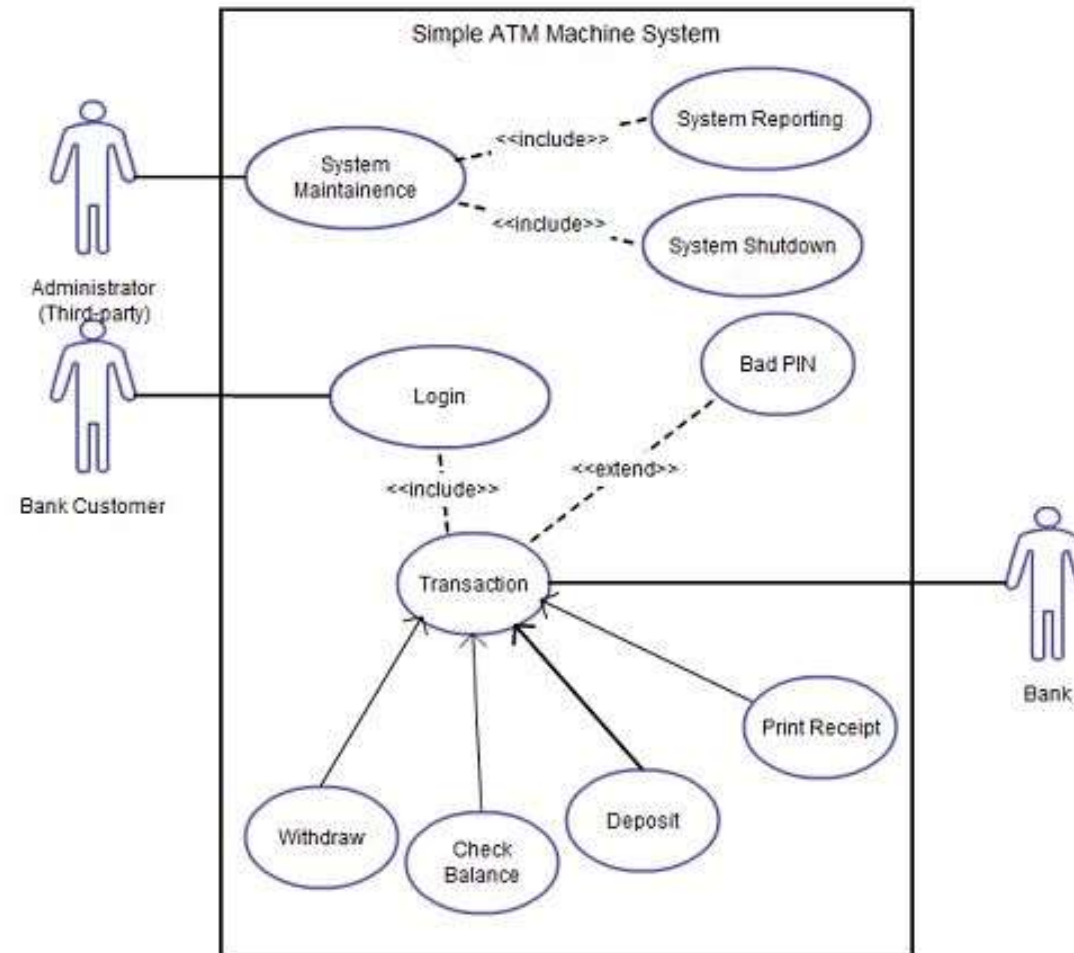


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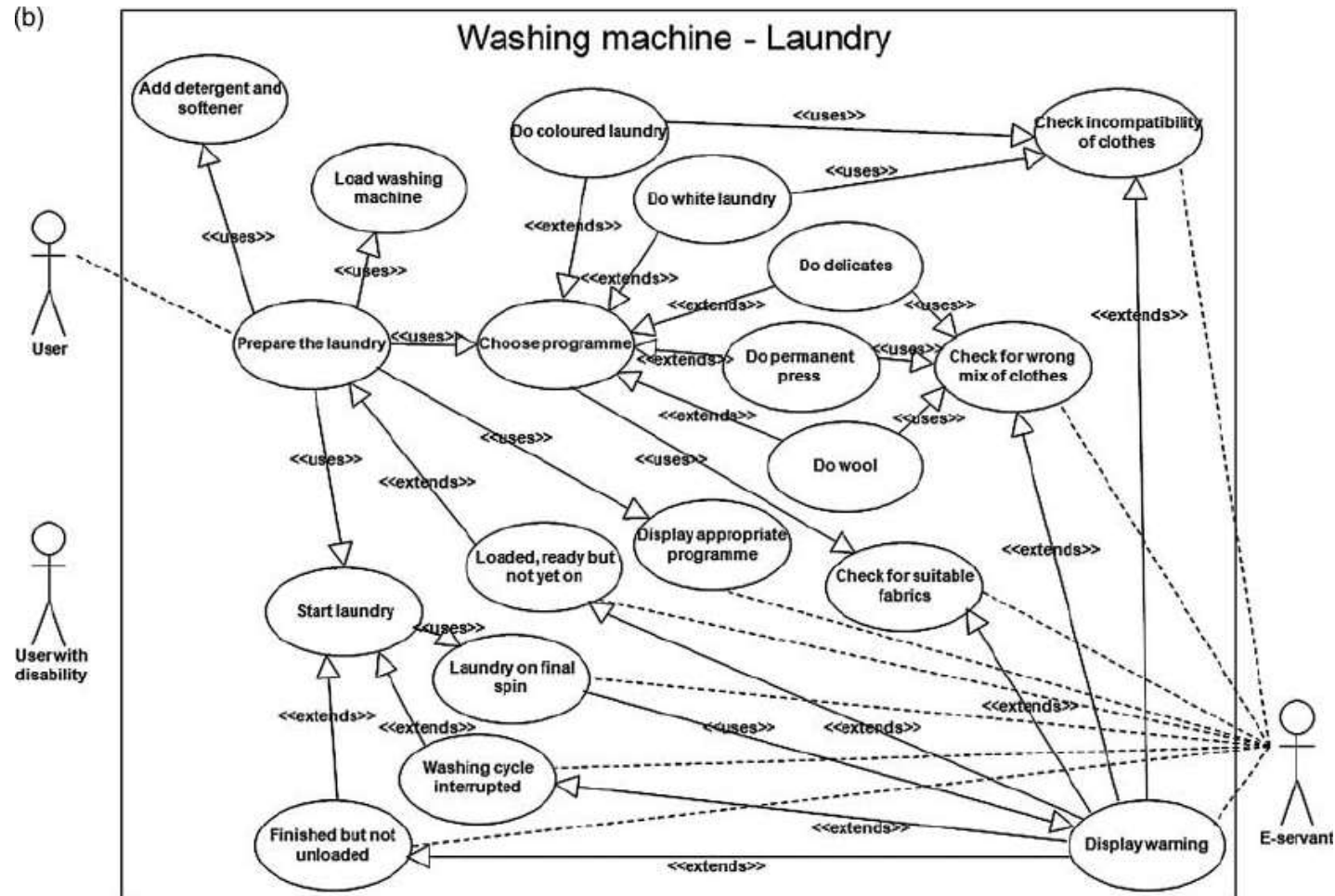
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USE CASE DIAGRAM EXAMPLE- ATM SYSTEM



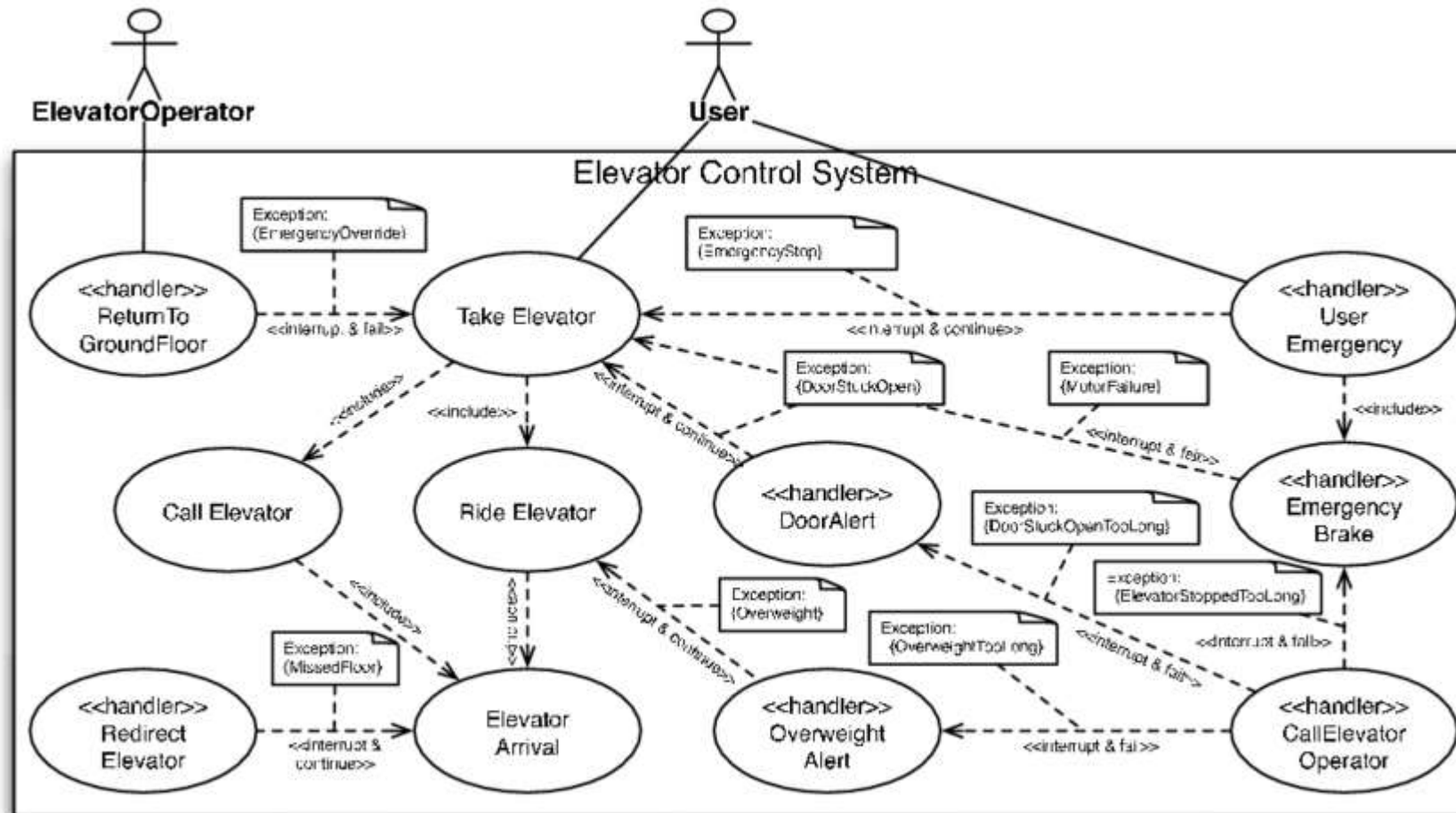
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USE CASE DIAGRAM EXAMPLE WASHING MACHINE LAUNDRY



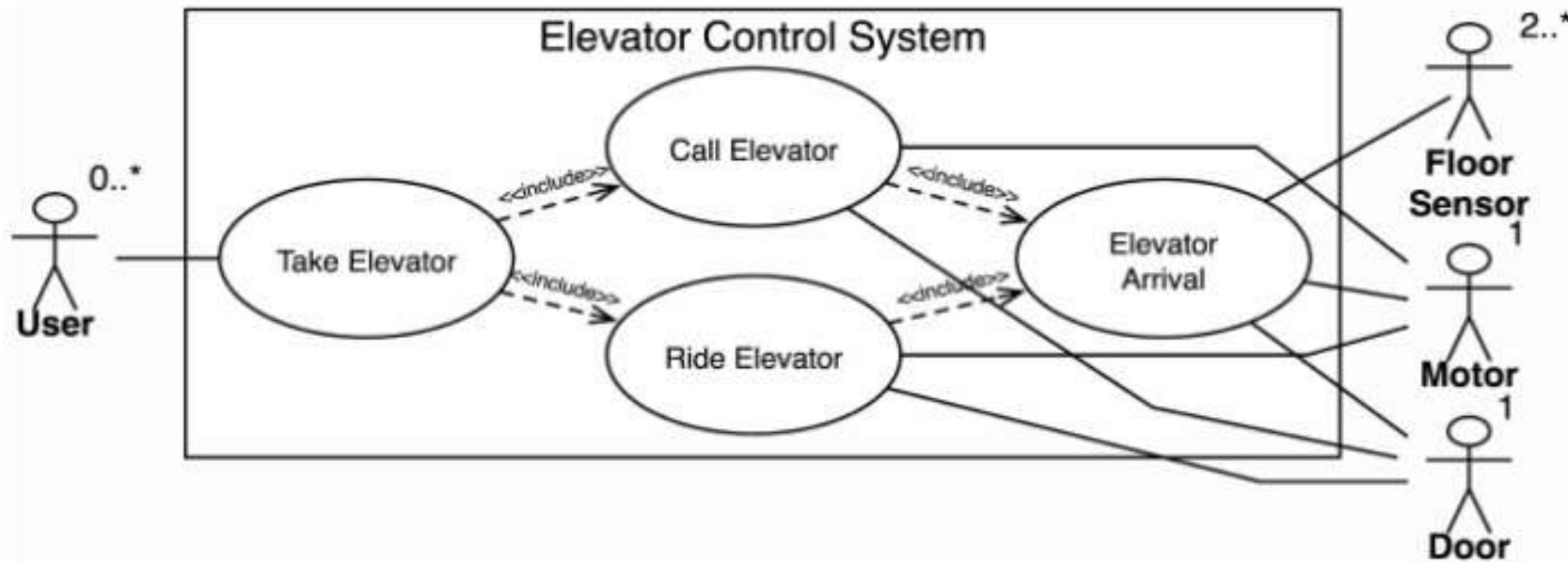
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USE CASE DIAGRAM EXAMPLE ELEVATOR SYSTEM



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USE CASE DIAGRAM EXAMPLE ELEVATOR SYSTEM



CLASS DIAGRAM

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CLASS DIAGRAM

- ❑ Class diagram- use to document software architecture
- ❑ It is used to refine the use case diagram
- ❑ Set of interrelated classes
- ❑ It contains methods and attributes for classes
- ❑ Classes can be "is-a" or "has-a" relationship

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CLASS DIAGRAM

- ❑ Class diagram - Static diagram, Structural diagram
- ❑ Class diagram - attributes and operations of a class
- ❑ It explains the constraints on the system
- ❑ It is a collection of classes, interfaces, associations, collaborations and constraints

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CLASS DIAGRAM

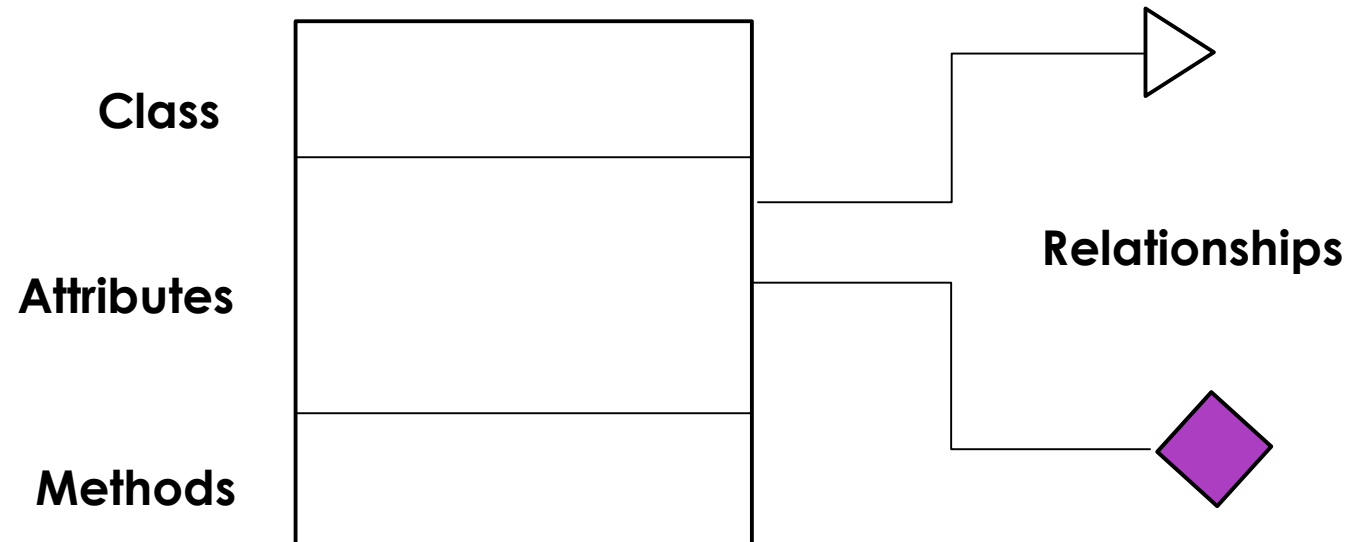
- ❑ The purpose of the class diagram can be summarized as –
 - Analysis and design of system
 - Describe responsibilities of a system
 - Conceptual modelling
 - Forward and reverse engineering

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UML NOTATIONS - CLASS

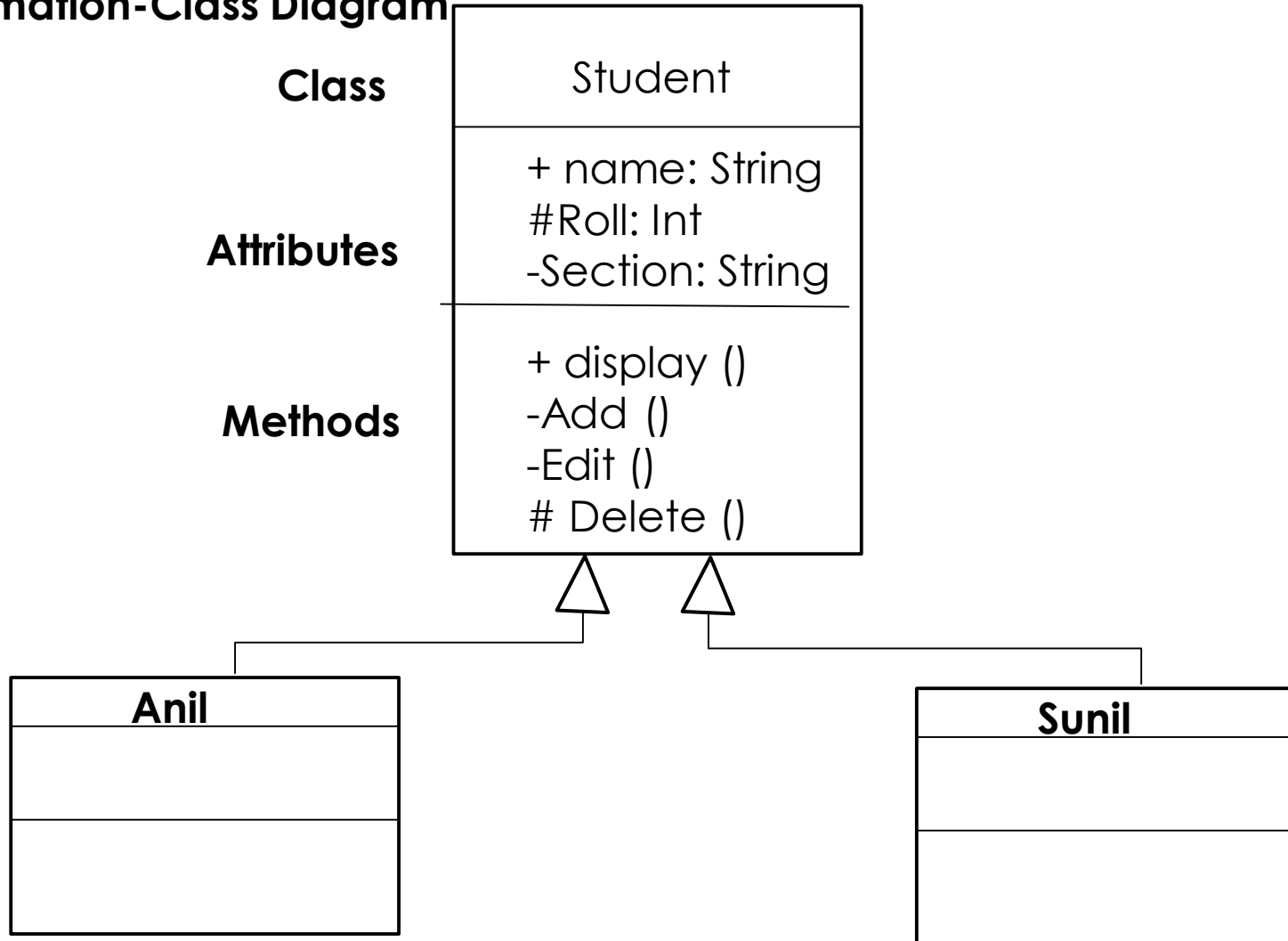
- ❑ **Classes are used to represent objects**
- ❑ **UML class diagram is divided into four parts.**
 - **Top section** - name the class.
 - **Second** - attributes of the class.
 - **Third** - operations or methods performed by the class.
 - **Fourth** - any additional components.

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Student Information-Class Diagram



Visibility

+ Public
- Private
protected
~ Package/default

Inheritance



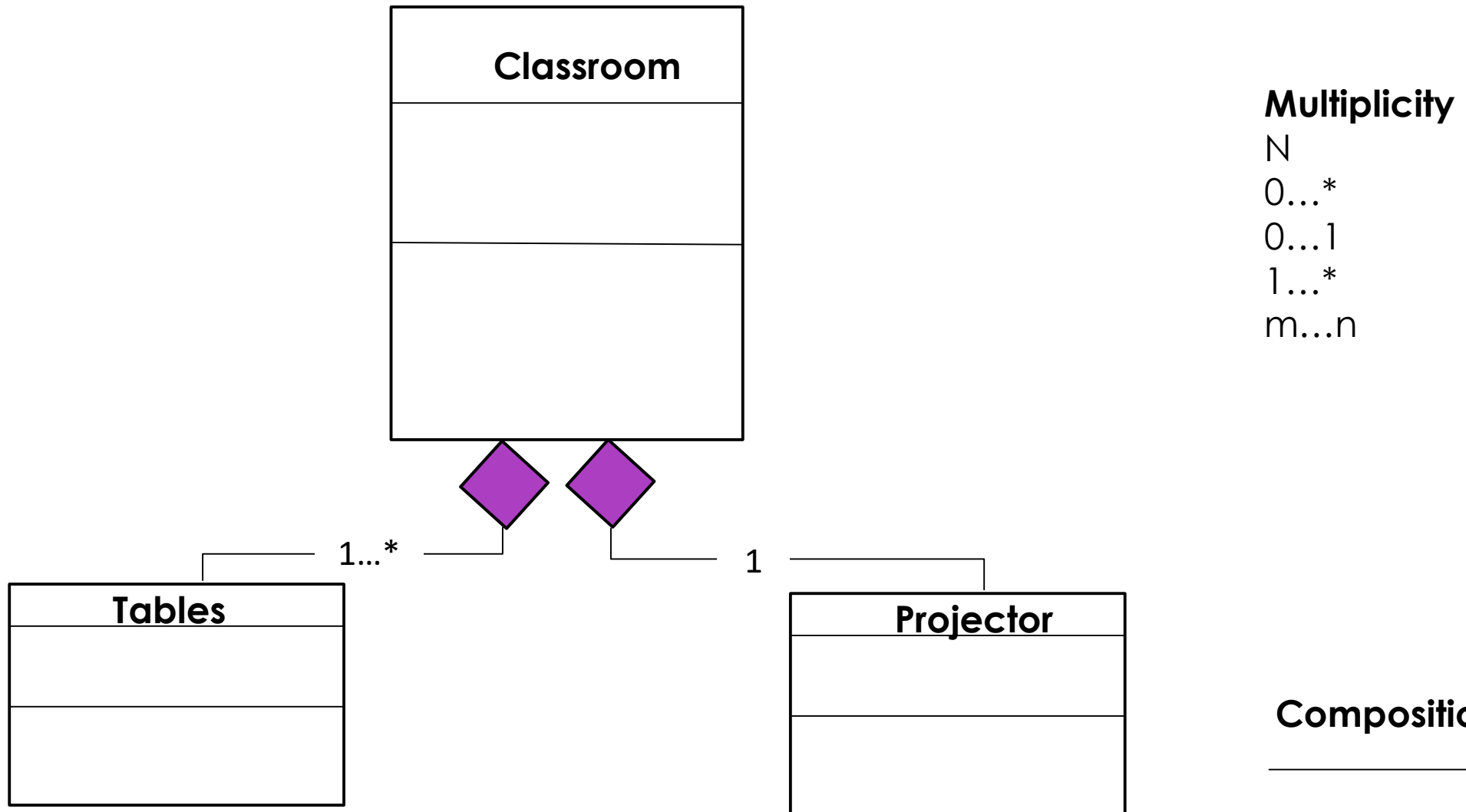
Association



Aggregation

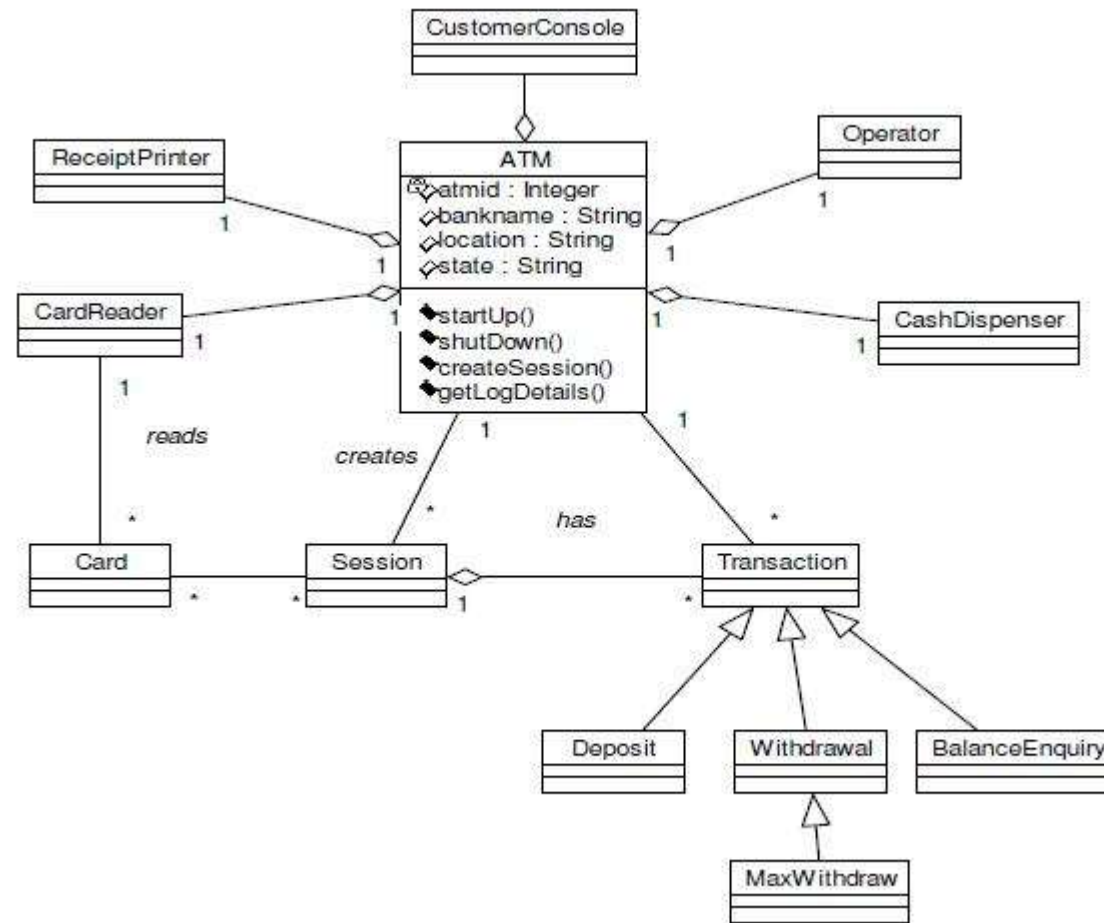


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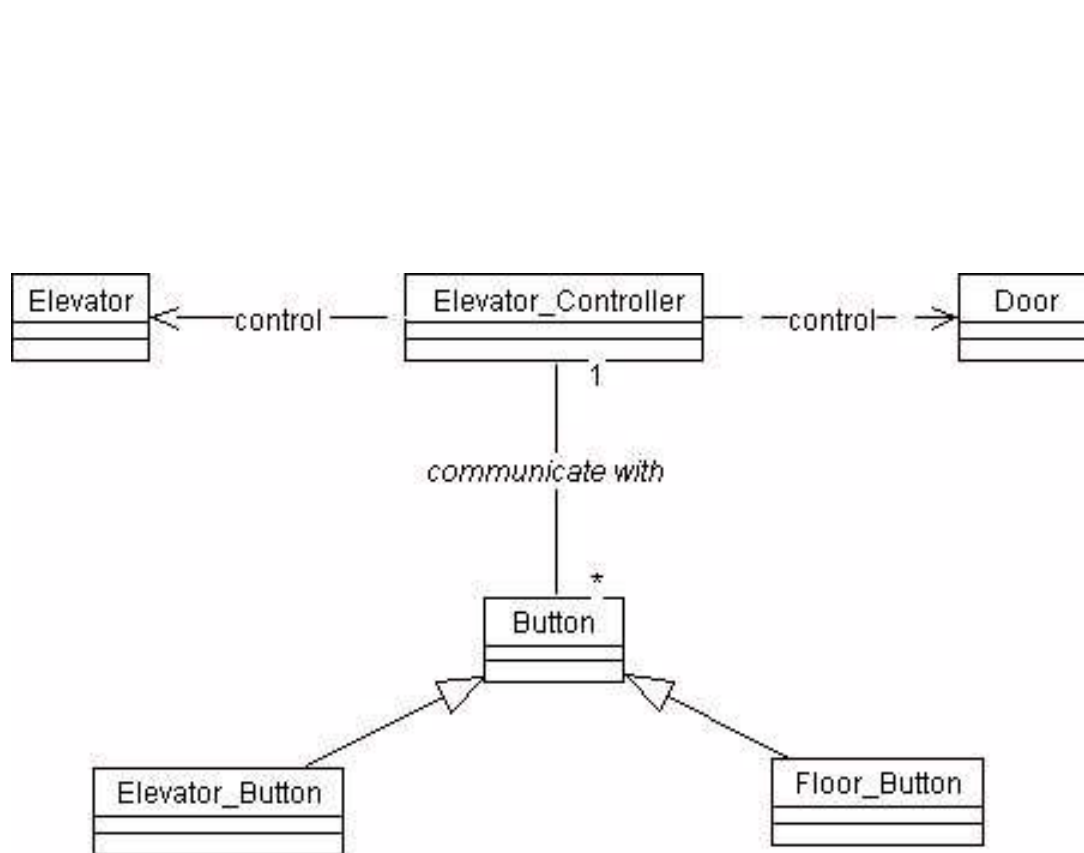
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CLASS DIAGRAM EXAMPLE- ATM SYSTEM

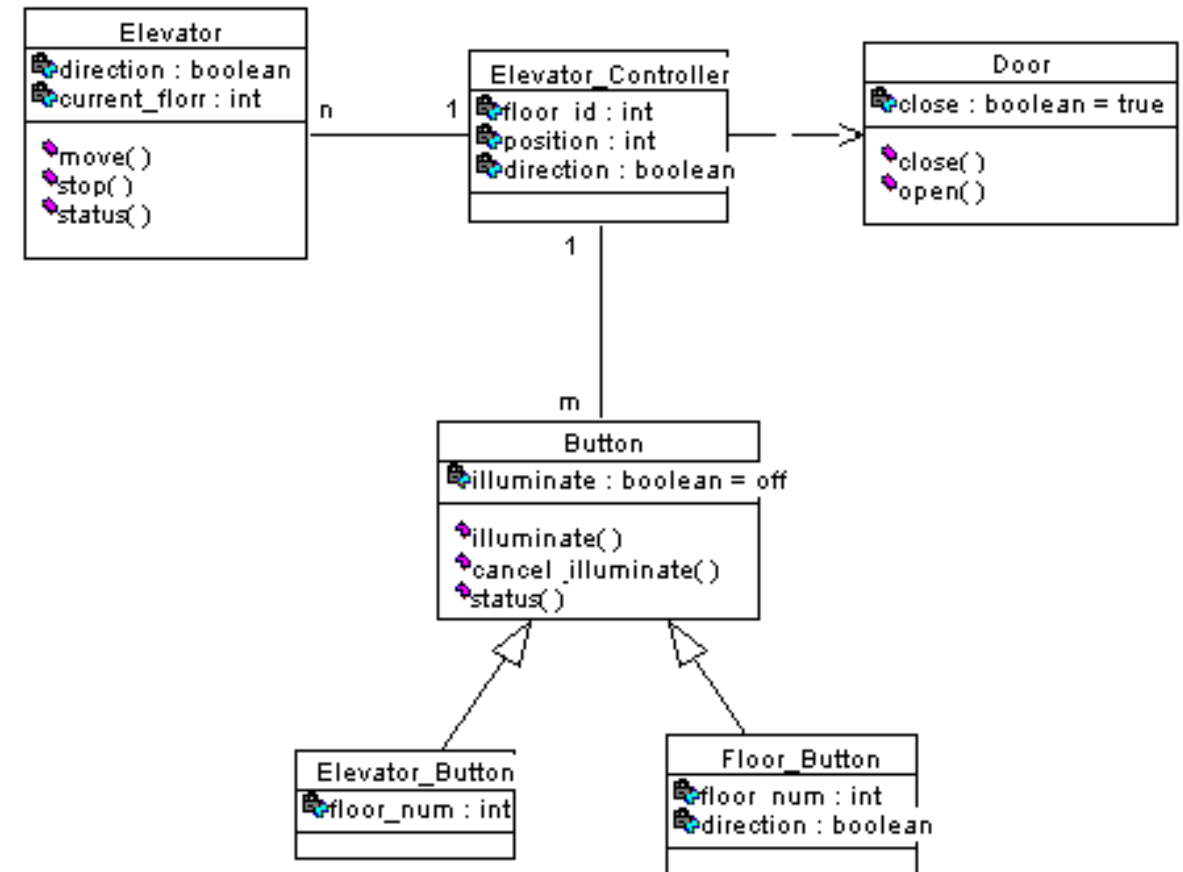


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CLASS DIAGRAM EXAMPLE ELEVATOR SYSTEM



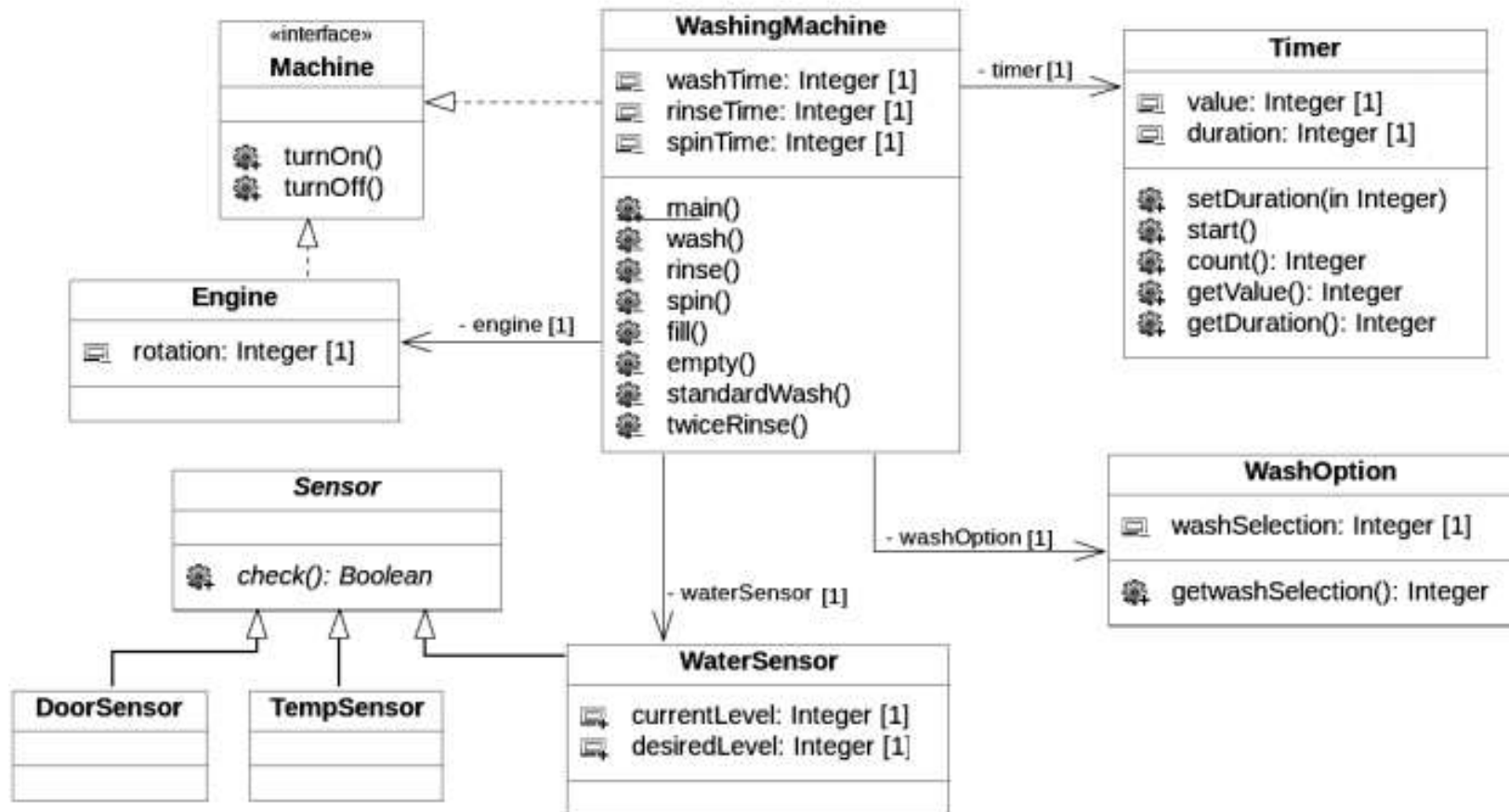
Simplified Class Diagram



Detailed Class Diagram

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CLASS DIAGRAM EXAMPLE WASHING MACHINE



SEQUENCE DIAGRAM

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SEQUENCE DIAGRAM

- ❑ Sequence diagram - interactions among the different elements in the model
- ❑ It describes the dynamic behavior of the system
- ❑ They explain the interaction between objects and collaborations
- ❑ It emphasizes on the time sequence of messages and structural organizations of the objects

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SEQUENCE DIAGRAM

- ❑ Sequence diagram shows the details of the use cases
- ❑ It shows the complete flow of information, function and operations of the system
- ❑ It is used in planning and understanding the functionality of existing and future scenario
- ❑ Time – vertical direction
- ❑ Header elements – Horizontal direction

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SEQUENCE DIAGRAM

- ❑ Sequence Diagrams are suitable for following scenarios:
 - ❑ Usage scenario – to determine how the system could be used
 - ❑ Method logic – to explore logic of any function, procedure or complex process
 - ❑ Service logic – ideal way to explain high level methods used by clients

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Actor



Object



Activation
Bar

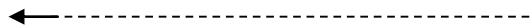
Bar



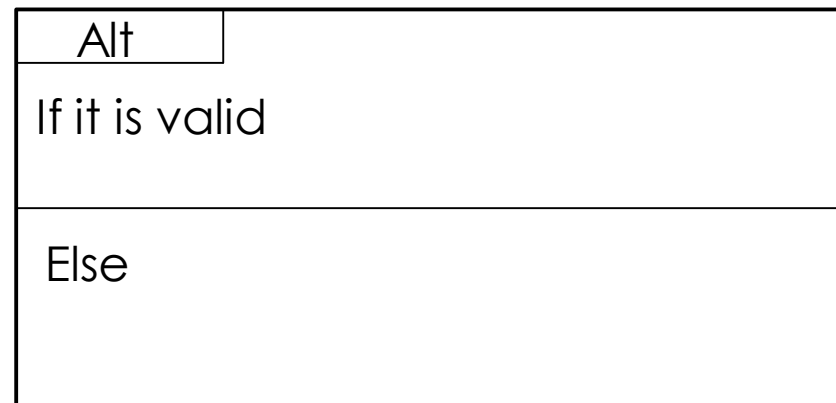
Synchronous message



Return message



Alternate frame



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SEQUENCE DIAGRAM EXAMPLE PRODUCT PURCHASE SYSTEM

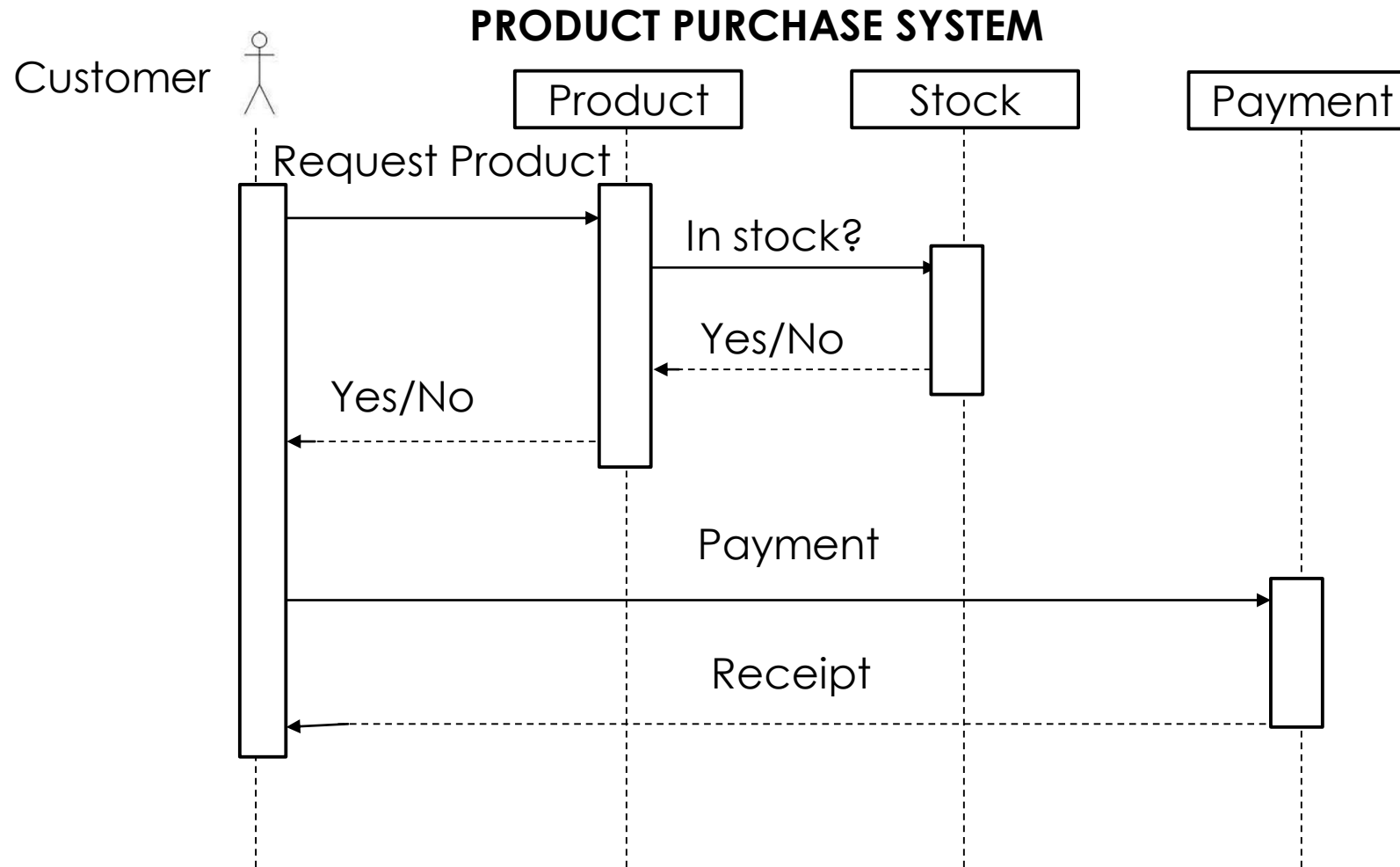
- ❑ In this sequence diagram we have four objects
 - Customer
 - Product
 - Stock
 - Payment
- ❑ The message starts from the top and flows to the bottom (waterfall manner)
- ❑ Dashed lines - duration for object
- ❑ Horizontal rectangles - activation of the object
- ❑ Messages sent - dark arrow and dark arrow head
- ❑ Return message - dotted arrow

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SEQUENCE DIAGRAM EXAMPLE - PRODUCT PURCHASE SYSTEM

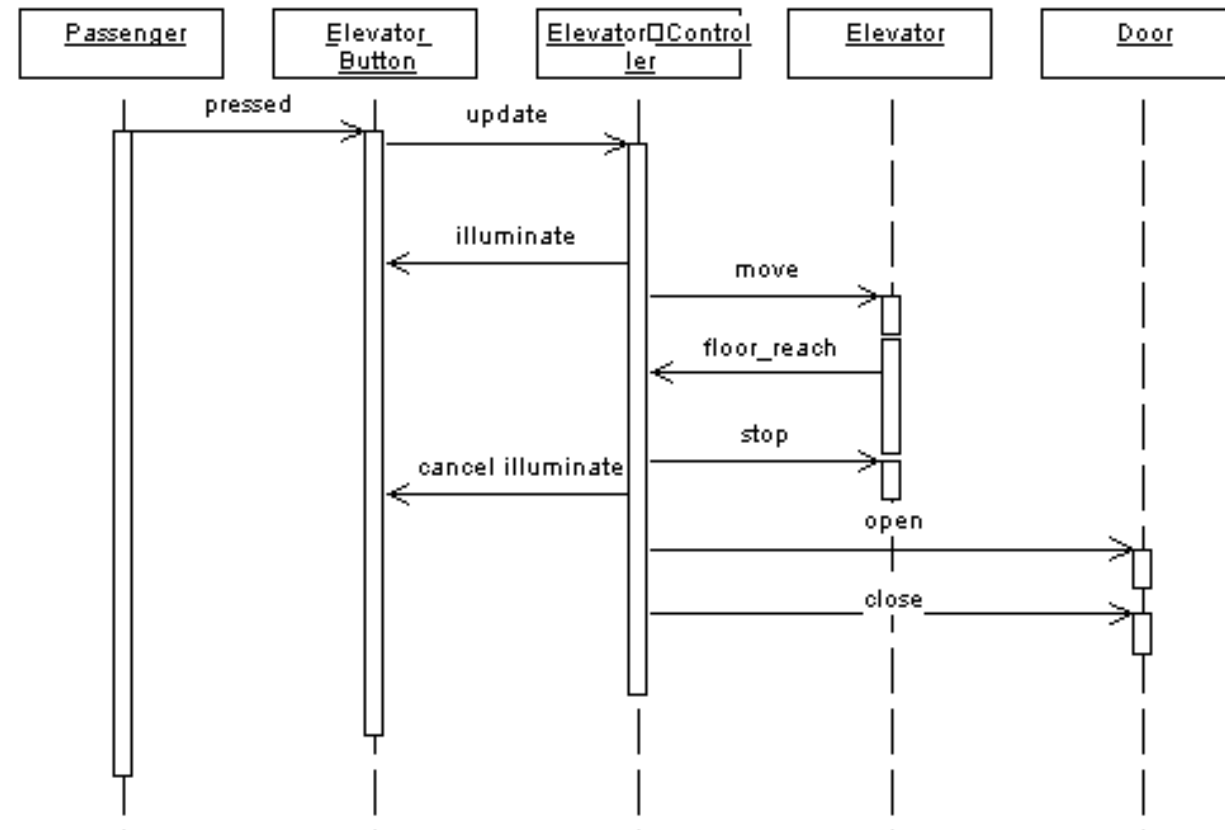
- Customer object - to the product object
- Product object -to the stock object
- Stock object - saying yes or No.
- Product object sends - customer object.
- Customer object -payment object to pay money.
- Payment object - receipt to the customer object.

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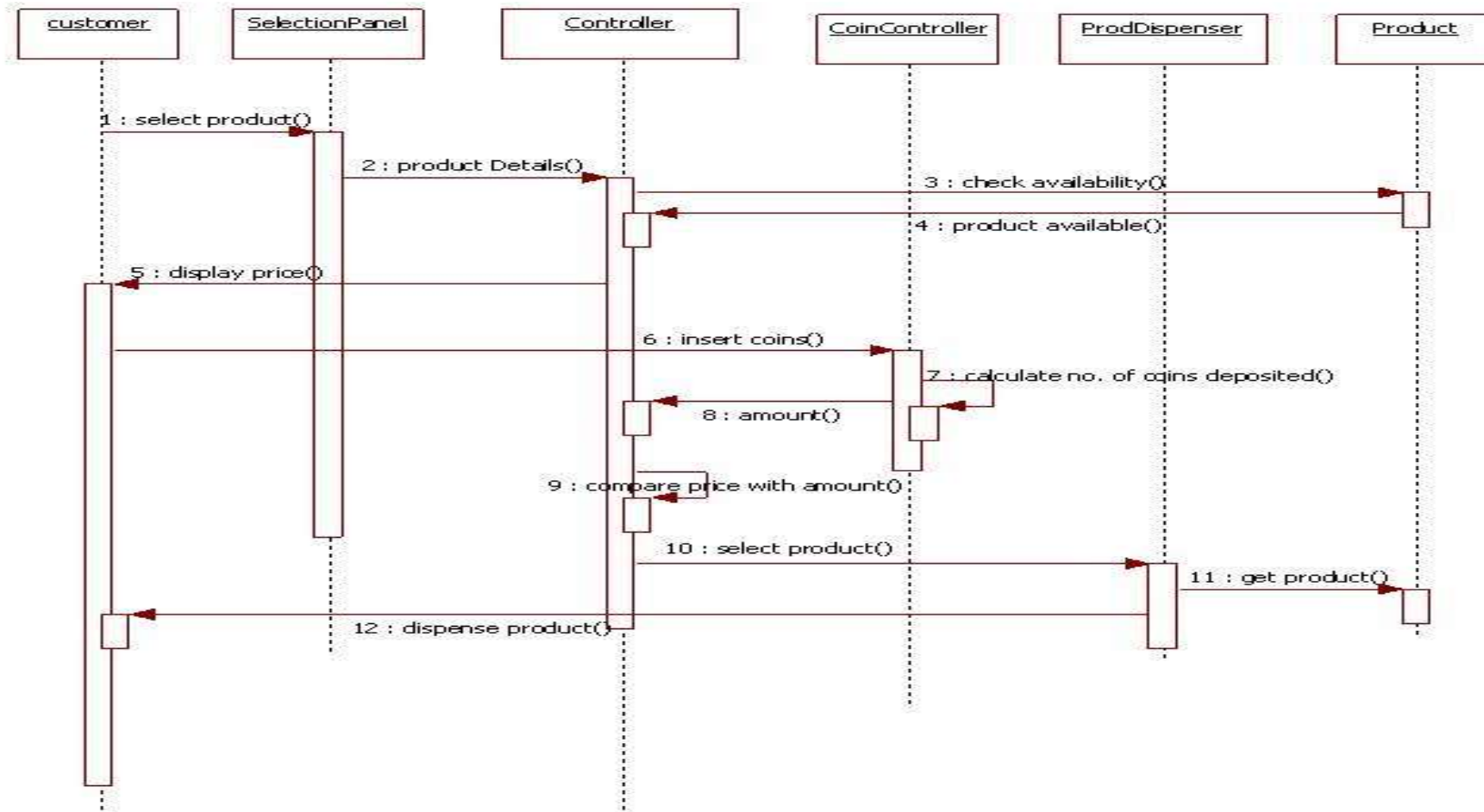
SEQUENCE DIAGRAM EXAMPLE ELEVATOR SYSTEM



Sequence Diagram for Serving Elevator Button

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SEQUENCE DIAGRAM EXAMPLE VENDING MACHINE



STATE DIAGRAM

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STATE DIAGRAM

- ☐ State diagram stores the status of object
- ☐ It shows the condition of system over a finite instance of time
- ☐ It's a behavioral diagram
- ☐ It also referred to as State machines and State-chart Diagrams.
- ☐ It explains the dynamic behavior of a class
- ☐ State diagrams can be used to understand the reaction of objects on receiving external stimuli

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STATE DIAGRAM

❑ Uses of state chart diagram:

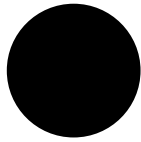
- We use it to depict event driven objects
- We use it for showing use cases in business context
- Shows the overall behavior of state machine
- We use it to model the dynamic behavior of the system .

❑ Steps to draw a state diagram:

- Identify the initial state and the final states.
- Identify the possible states in which the object can exist
- Label the events which trigger these transitions.

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STATE DIAGRAM SYMBOL NOTATIONS



Initial state



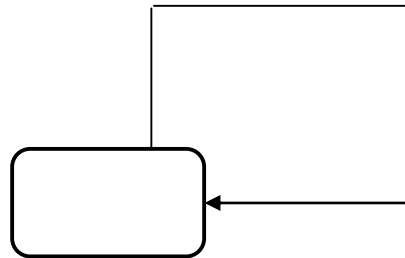
Transition Symbol



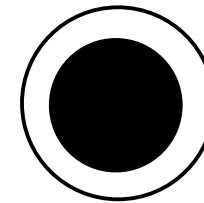
State



Composite State



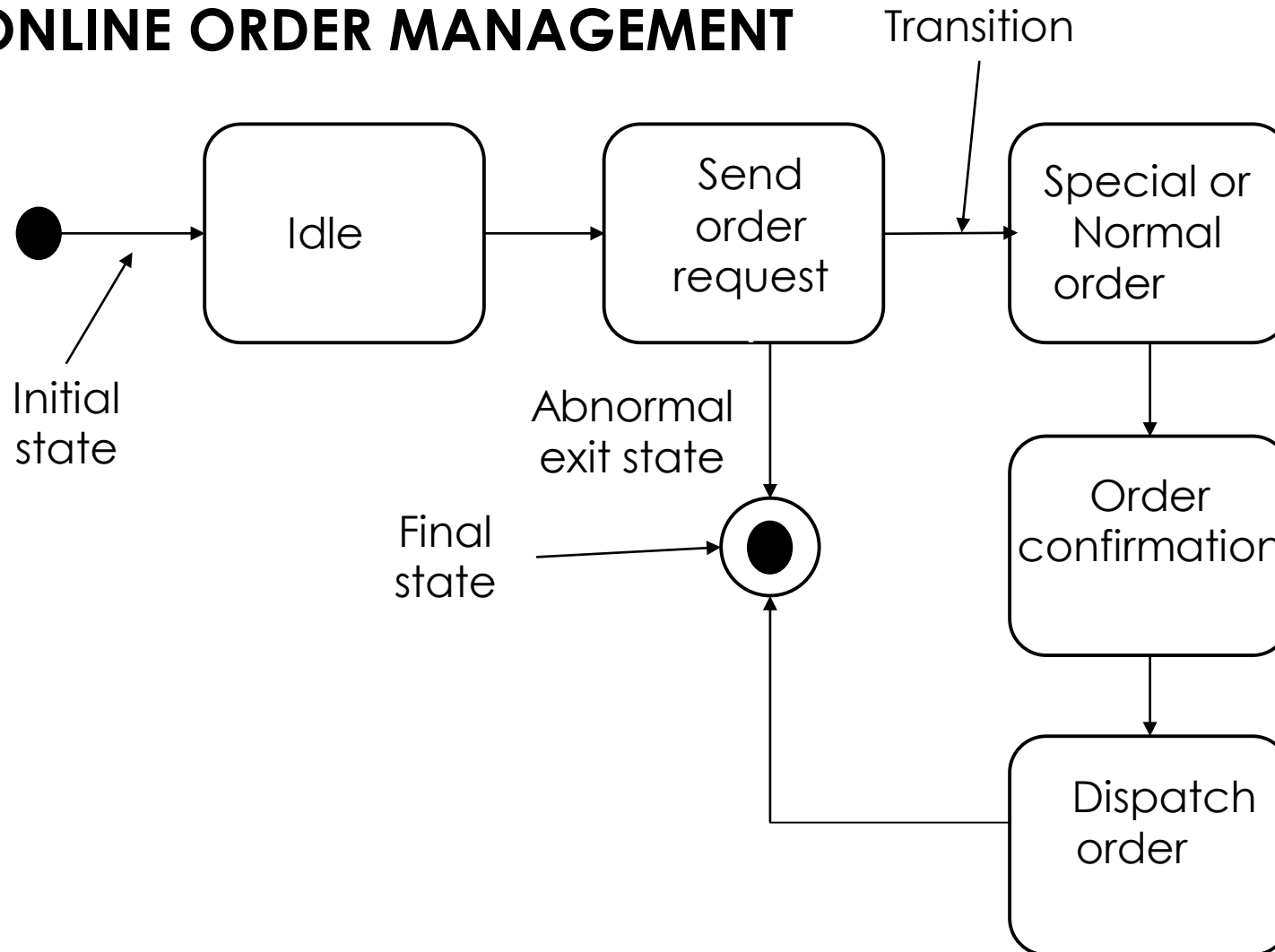
Self Transition



Final state

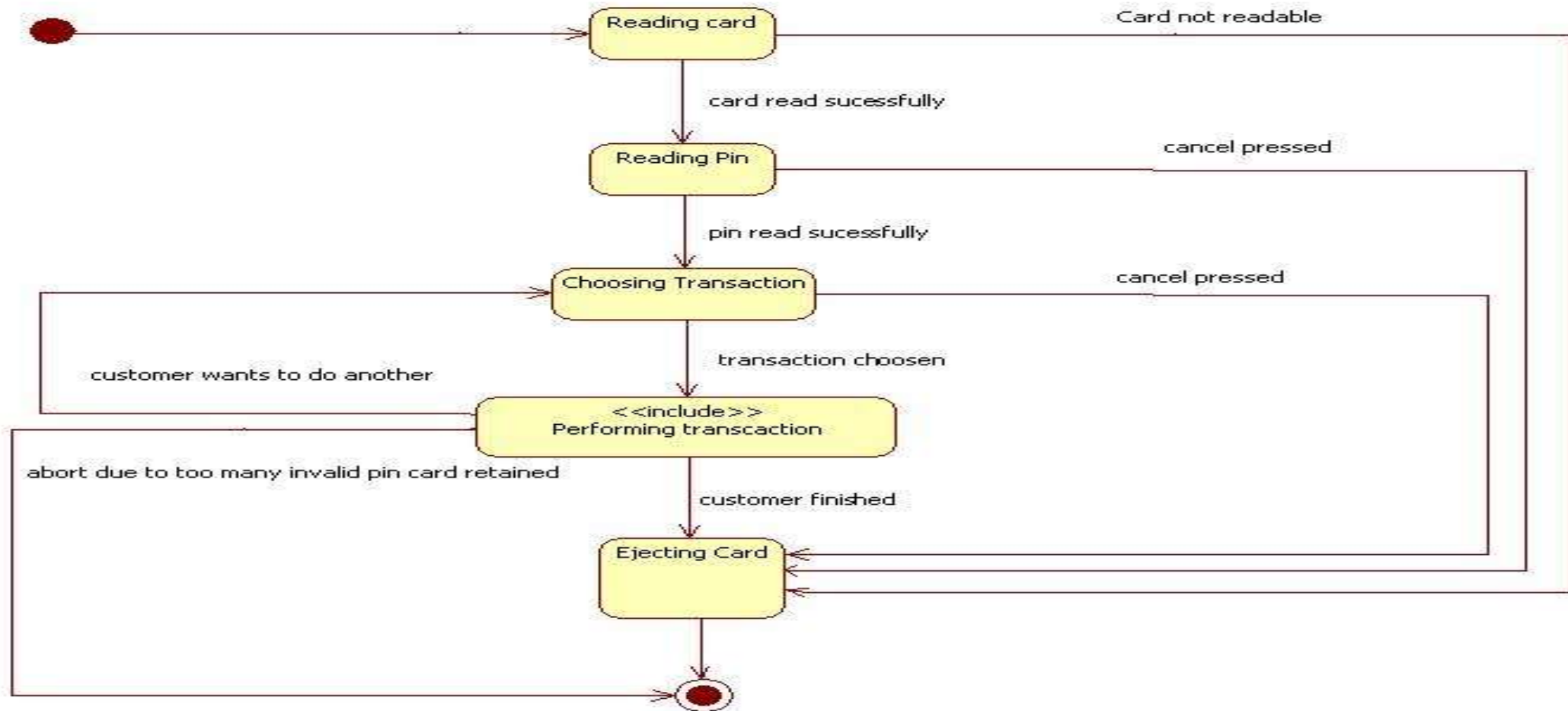
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ONLINE ORDER MANAGEMENT



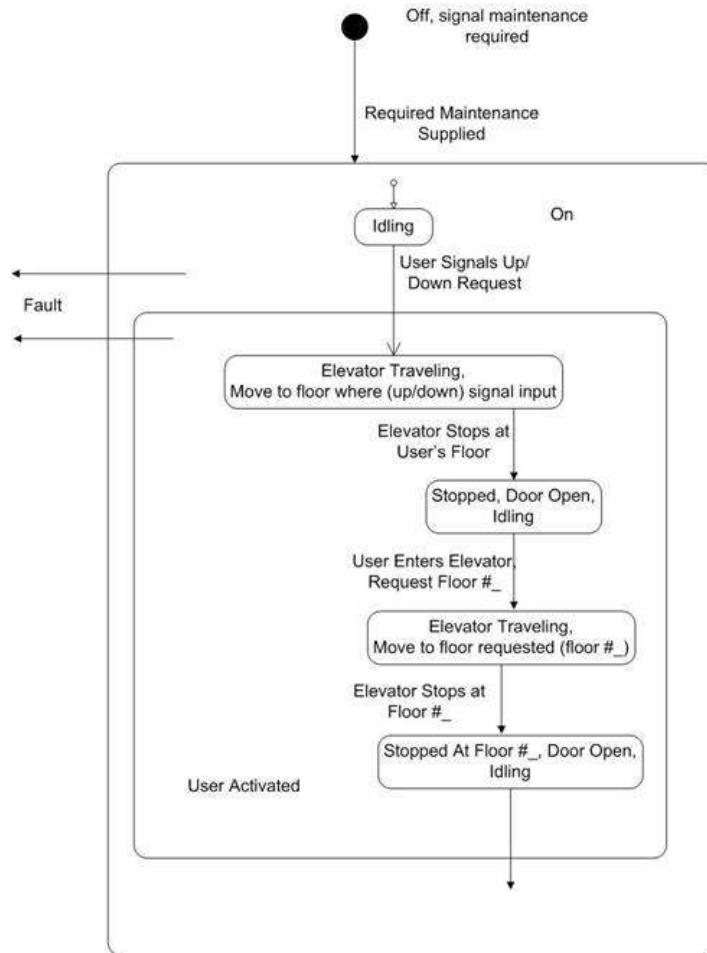
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STATE DIAGRAM EXAMPLE ATM MACHINE

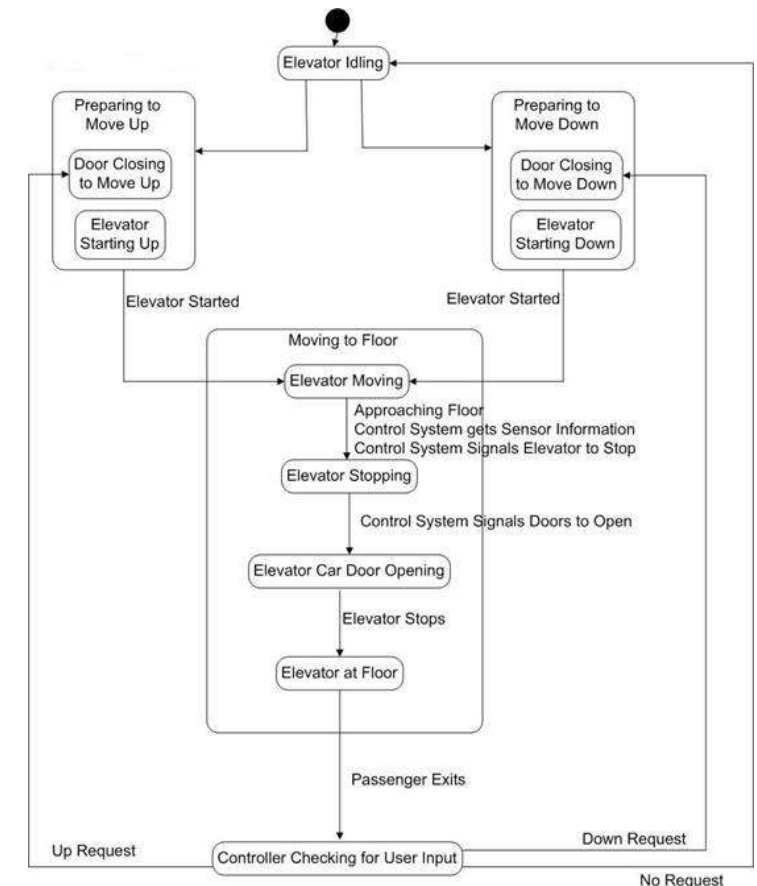


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STATE DIAGRAM EXAMPLE ELEVATOR SYSTEM



Statechart for
"User"
Behavior



Statechart
for
"Elevator
Control
System"

ACTIVITY DIAGRAM

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ACTIVITY DIAGRAM

- ❑ An activity diagram - behavioral diagram
- ❑ It shows the control flow from start to finish point including other decisions paths that is being executed
- ❑ It can explain both sequential and concurrent processing

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ACTIVITY DIAGRAM

- ❑ Uses of Activity chart diagram:
 - Modelling work flow by using activities
 - Modelling the logic of algorithm
 - Shows the workflow process between users and system
 - Understanding system's functionalities.
 - Investigating business requirements at a later stage.

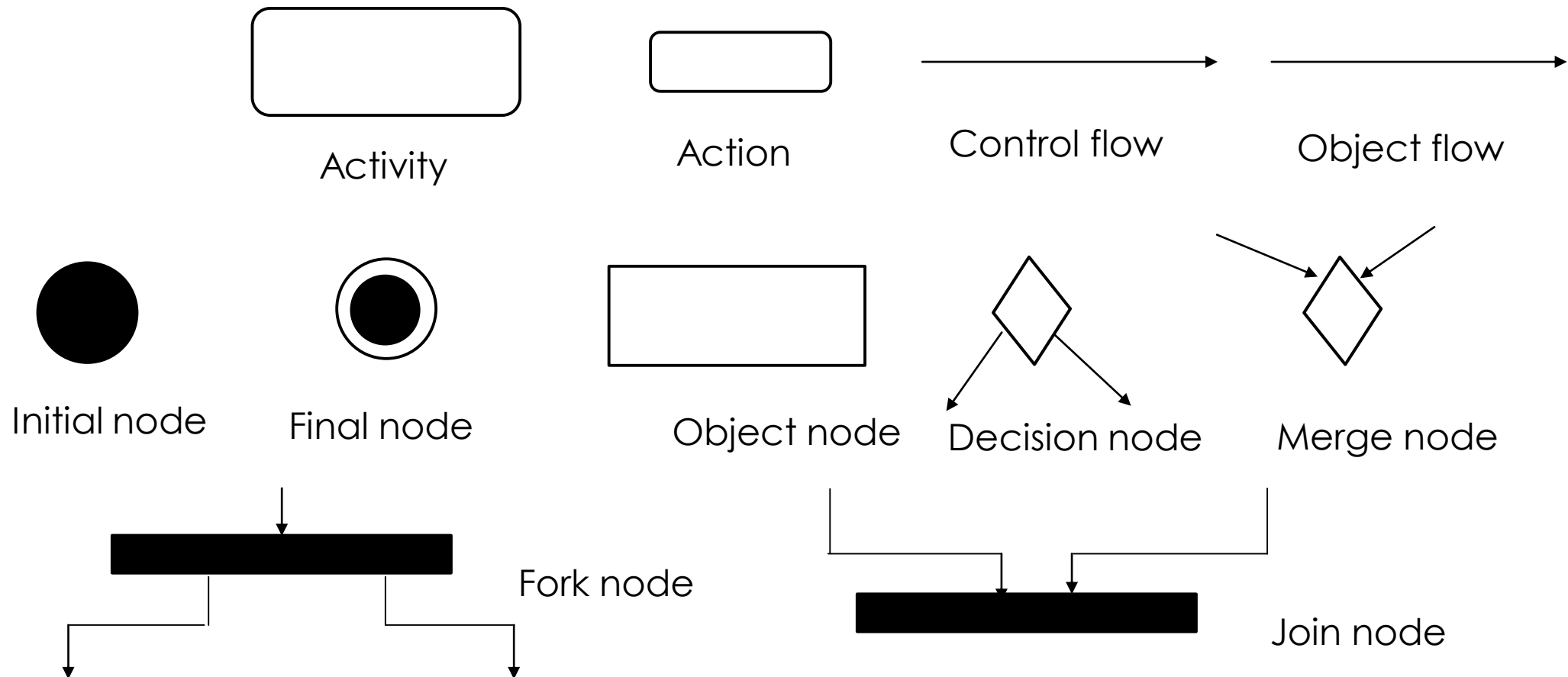
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ACTIVITY DIAGRAM

- ❑ Steps to draw a activity diagram:
 - Identify the initial state and the final states.
 - Identify the intermediate activities needed to reach the final state from the initial state.
 - Identify the constraints causing the system to change control flow.
 - Draw the diagram with appropriate notations.

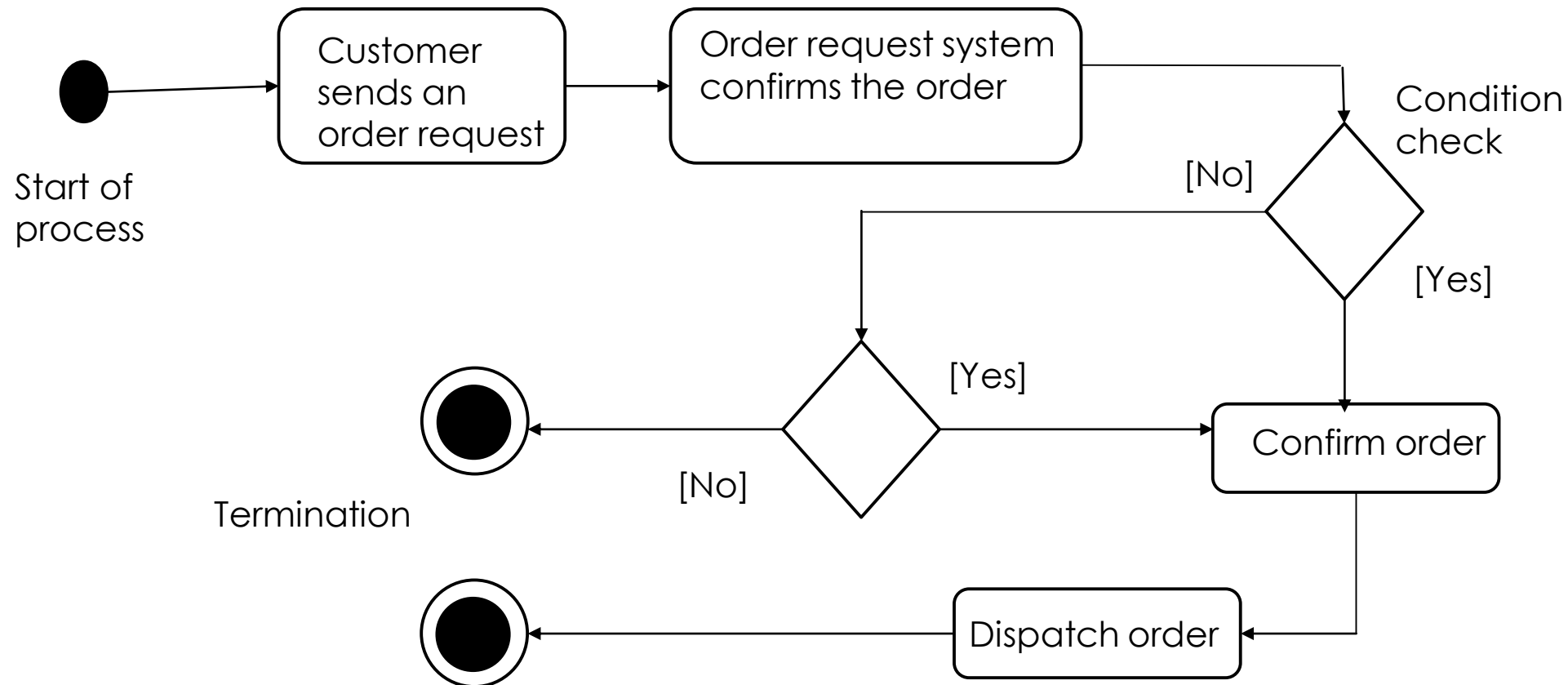
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ACTIVITY DIAGRAM SYMBOL NOTATIONS



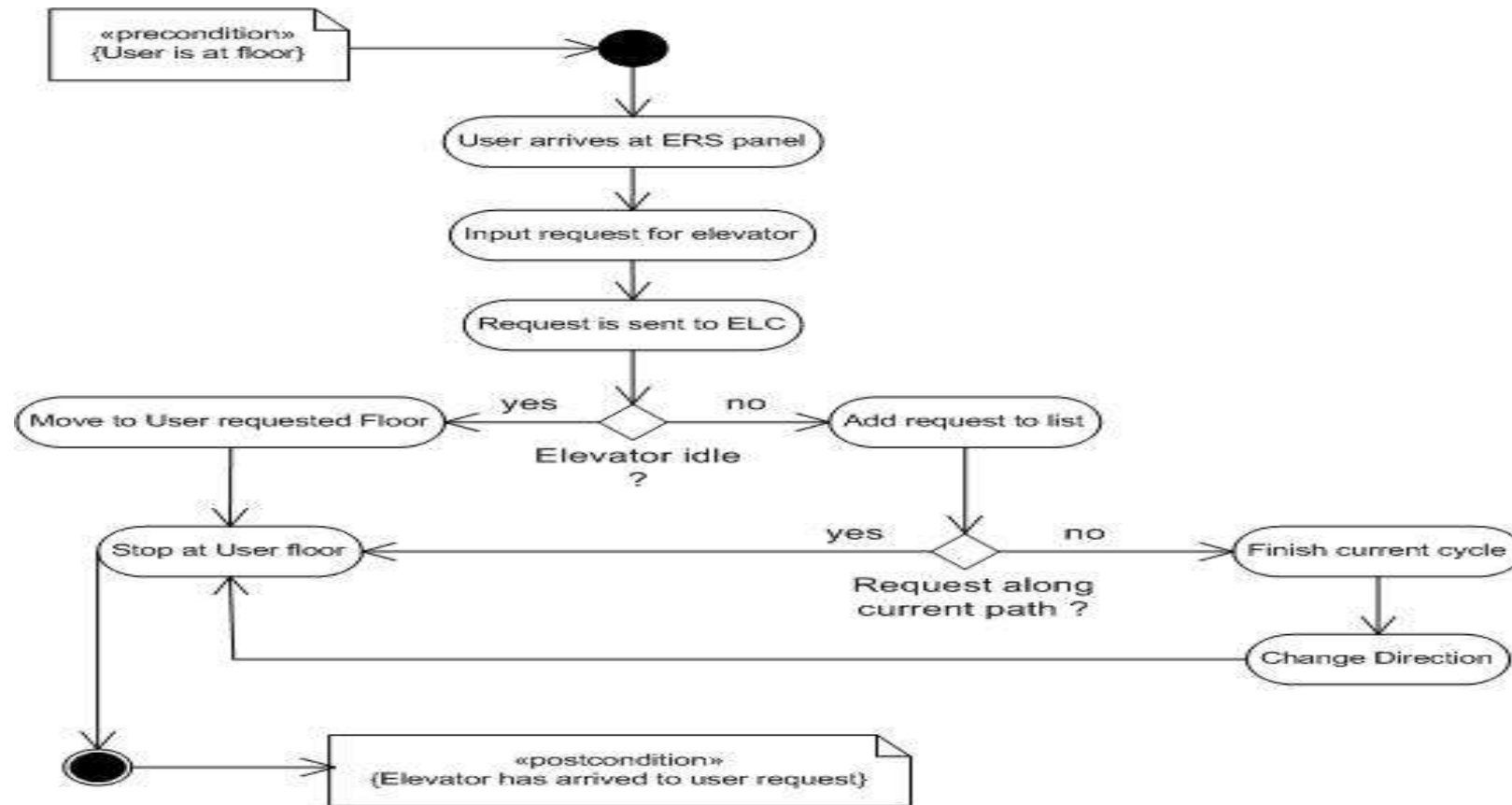
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Activity Diagram of an Order Management System



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ACTIVITY DIAGRAM EXAMPLE ELEVATOR SYSTEM



ELC – Elevator Logic Control
ERS – Elevator Request Service

Request Elevator Activity

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ACTIVITY DIAGRAM EXAMPLE VENDING MACHINE

